Boolean Algebra

Yuna Kang

1 Introduction

In 1874, Alexander Graham Bell found the first telephone successful. In 1993, the first smartphone was launched, the IBM Simon, about 8 inches high and weighing 18 ounces. Only about 30 years later, the average size of cell phones is pocket-sized, only about 6.3 inches high, and weighing less than 8 ounces. The abrupt advancement of phones that are used in real life was enabled by computer science and mathematics, specifically Boolean Algebra.

2 Boolean Algebra

Phones and other electronic devices consist of hundreds of circuits that store data. They are formed by a series of microscopic *logic gates*. An input will signal the output to turn off or on, or 0 or 1 respectively. Since it has only two possible outputs, it is called a binary system.

Boolean algebra was invented by George Boole, an English mathematician and philosopher, when he published The Mathematical Analysis of Logic (1847). Boolean algebra simplifies operations with variables that are either true (1) or false (0) by using properties, which results in a faster and smaller device.

2.1 Operations

Operations in Boolean Algebra are logical processes from a certain input that yields a certain output. The common fundamental mathematics have four operations: addition, subtraction, multiplication, and division. While there are more operations such as the NAND, XOR, NOR operations, there are three major operations in Boolean Algebra: the NOT operation, AND operation, and OR operation.

2.1.1 NOT Operation

The NOT Operation, denoted x' or $\neg x$, inverts the boolean digit by using the formula $1 - x$. In terms of mathematical logic, it is known as negation. The operation states that for an input, the output is the inverse of the input.

2.1.2 AND Operation

The AND Operation, denoted $x \cdot y$ or $x \wedge y$, is similar to multiplication (mod 2) or conjunction in mathematical logic. The operation states that only if both inputs are true, the output is true. In other words, if the given statements are "I am a student, and I enjoy school", then both statements "I am a student" and "I enjoy school" would be true.

2.1.3 OR Operation

OR Operation, denoted $x + y$ or $x \vee y$, is similar to addition mod 2 or disjunction in mathematical logic. The operation states that only if both inputs are false, the output is false. For example, if the given statements are "It is snowing, or it is raining", then at least one of the statements "It is snowing" or "It is raining" is true.

However, sometimes there may be a given statement "I am in middle school, or I am in high school." Since a person cannot be both in middle and high school, only one of the statements are true. This is known as the XOR Operation, or the exclusive OR. OR Operations are generally inclusive and are true when both statements are true.

2.2 Truth Tables

The truth table is a way of showing the operation's outcomes by listing out all possible values either true or false for every combination possible. If there are n variables, the number of combinations for the truth table is n^2 since the variables can only be true or false.

Table 1: Boolean Operations

x	x'
0	1
1	0

(a) NOT Operation

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

(b) AND Operation

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

(c) OR Operation

2.3 Logic Gates

Logic gates is a device that does a Boolean operation. The millions of logic circuits inside an electronic device can be mapped out by logic gates.

3 Laws of Boolean Algebra

While Boolean operations relates the inputs to the outputs, there are also laws of Boolean algebra that can simply these logic gates. Simplifying these complex logic gates with the

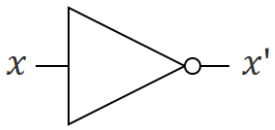


Figure 1: NOT Gate

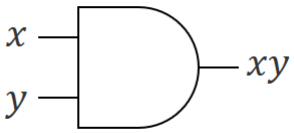


Figure 2: AND Gate

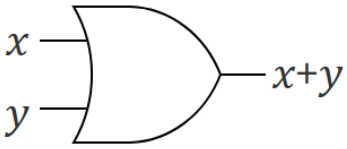


Figure 3: OR Gate

same input and output combination, can help reduce the size of the electronic device while also improving the speed of the device.

Law	AND	OR
Annulment	$a \cdot 0 = 0$	$a + 1 = 1$
Identity	$a \cdot 1 = a$	$a + 0 = a$
Idempotent	$a \cdot a = a$	$a + a = a$
Complement	$a \cdot a' = 0$	$a + a' = 1$
DeMorgan's Theorem	$(ab)' = a' + b'$	$(a + b)' = a'b'$
Commutative	$ab = ba$	$a + b = b + a$
Associative	$(ab)c = a(bc)$	$(a + b) + c = a + (b + c)$
Distributive	$a + bc = (a + b)(a + c)$	$a(b + c) = ab + ac$
Absorption	$a(a + b) = a$	$a + ab = a$

Table 2: Laws of Boolean Algebra

4 The Binary System

It is commonly said that computers "only understand zeros and ones," and indeed information of all kinds from numerical, text, colors, images, colors, letters, to even audio is encoded using these two symbols. In other words, computers rely on the binary number system, a base 2 representation using digits 0 and 1. In contrast, humans generally use the decimal, or base 10, number system, with ten digits ranging from 0 to 9. Despite its minimal alphabet, binary provides a powerful and efficient way to represent and manipulate information electronically.

4.1 Decimal System to Binary System

As mentioned before, computers uses a binary system to calculate numbers, while humans use the decimal system. However, the fundamental for a base 2 system and a base 10 system are very similar.

Take the number 72 in base 10, for example. 72 in the decimal system can be represented as $(7 \times 10^1) + (2 \times 10^0)$. Each numbers from 0 through 9 is being multiplied by 10^n, where n

is an integer. In order to find 72 in the binary system would be to find each number from 0 to 1 being multiplied by 2^n, where n is an integer. 72 can be expressed as $(1 \times 2^6) + (1 \times 2^3)$. So in the binary system, the number 72 is 01001000.

4.2 Binary System Representation

Converting the decimal system into the binary system can be used to represent numbers in the computer.

Computers uses the binary system as well as letters, audio, colors, etc. For example, for the binary codes corresponding to each letter, they use Unicode or ASCII (American Standard Code for Information Interchange). In fact, 01001000 01100101 01101100 01101100 01101111 00100001 in binary code means "Hello!"

All kinds of information are assigned a specific combination of zeros and ones. To prevent a misinterpretation of differentiating 01001000 as the number "72" instead of the letter "H", computers have programs finds if a specific code is referring to a number or a letter using context.

5 Conclusion

Electronics uses only zeros and ones, or the binary system. Boolean algebra also uses a binary system and has been fundamental to the creation and evolution of the digital technologies that make up the necessities of our daily life and will continue to influence the future. While we often use our phones and computers, without noticing the abstract mathematics beneath them, it is worth appreciating how Boolean logic helped make modern computing possible.